Experiment to demonstrate negative geotropism in aerial stem

The Clinostat has a rotating pot like container mounted on an axis rod. A potted plant is fitted horizontally on the Clinostat and rotated slowly which completely eliminates gravity as all the sides of the plant are equally stimulated. If the rotation of the Clinostat is stopped for a considerable period of time then the tip of the stem is observed to curve and grow upwards this proves that the stem tip is negatively geotropic (Figure 15.15).

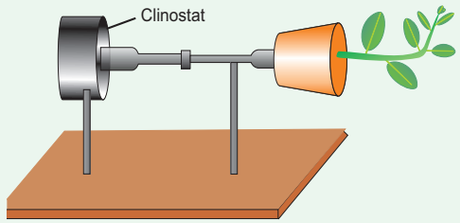


Figure 15.15 Clinostat

Experiment to demonstrate positive phototropism in shoot tips

A darkened black box is taken having a small window on one side. A well-watered potted plant is placed inside the box. This is referred to as a phototropic chamber or heliotropic chamber. If the window is kept closed for about 24 hours the plant shows normal growth. If the window is kept opened, it is found after two days that the shoot tip bends and grows towards light proving that it is positively phototropic (Figure 15.16).



Figure15.16 Experiment to demonstrate Photropism

while others such as roots and rhizoids which move away from the stimulus of light are called **negatively phototropic**.

b. Nastic Movements

When growth movements occur in response to an external stimulus which is not unidirectional but diffused, they are called nastic or paratonic movements of variation. Paratonic variation movements are determined by some external stimuli, light, temperature, chemicals and touch. They are:

1. Nyctinastic movement (or) sleep movement

The diurnal (change in day-night) movements of leaves and flowers of some species which take up sleep position at night are called **nyctinastic movements**. They are caused by relative changes in cell size on the opposite sides of the leaf base called **pulvinus**. The movements are attributed to the amount of auxin and K^1 ions. The entry of water to the lower side of the pulvinus causes the leaves to stand erect and exit of water causes them to drop. They are of two types:

i. Photonasty

The nastic movement caused in response to light is called **photonasty** or **photonastic movement.** The opening of leaves and flowers during daytime and their close at night is an example.

ii. Thermonasty

The nastic movement taking place in response to temperature is called **thermonasty** or **thermonastic movement**. In *Crocus* the flowers open at high temperature and close at low temperature.

2. Seismonastic movement

This means a response to shaking. The best example is *Mimosa pudica* (Touch-me-not plant) which is a sensitive plant (Figure 15.17). Such plants respond to stimuli such as touch, blow or metallic shock by folding their leaflets and lowering their leaves .This effect is caused by the movement of water in



Figure 15.17: *Mimosa pudica* showing response to touch

XIth_Botany_Chapter_15_EM.indd 317 5/17/2020 9:21:56 PM